
Cortical Decoders Trained on Other Individuals Age Slower Than Your Own

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Abstract

Cortical representations drift; a decoder fit to one animal’s neural activity slowly loses accuracy on that animal over days, even with unchanged behavior. The standard solution is periodic recalibration on fresh data from the subject; we report a different form of robustness. On the open BraiDyn-BC cohort (25 mice, cortex-wide widefield $\Delta F/F$ parcellated into 44 Allen-atlas regions, a ~ 15 -day operant protocol), we decode four task events and compare, per mouse, how much accuracy is lost over two weeks by (i) the mouse’s own early-week decoder and (ii) a decoder pooled over the other 24 mice’s early-week data. Trials recorded in one session share imaging conditions, hemodynamic state and slow arousal, so every within-subject estimate here holds out whole sessions rather than trials; splitting at the trial level inflates the personal decoder, and therefore the asymmetry itself, by up to 0.013 AUC. Under the session-held-out protocol the pooled decoder ages less: the personal-minus-pooled asymmetry is +0.017 AUC ($p = 5 \times 10^{-5}$; the own decoder ages quicker in 73 of 100 mouse–target pairs). The effect is carried by two of the four events, lick (+0.020, $p = 0.002$) and reward (+0.029, $p = 0.0005$), while tone is marginal (+0.015, $p = 0.08$) and lever pull is null (+0.003, $p = 0.37$). A day-by-day regression across the full 15-day protocol independently finds a narrowing personal-versus-pooled gap for exactly those two events, so two analyses with different estimators agree on which events carry the effect. A count-matched control isolates the cause: holding the number of training events fixed, the subject’s own decoder ages more than one trained on a single other mouse for lick (+0.007), reward (+0.017) and tone (+0.006). As both see equal data this is not a data-volume effect; it follows from not overfitting the subject’s own drift-prone idiosyncrasies. Training on many other mice rather than one lowers aging further on all four events. Population pooling is thus a drift-mitigation strategy in its own right: the early-week cost of decoding across individuals is repaid as temporal stability. Code and a streamed-feature pipeline are released.

1 Introduction

Representational drift, the reorganization of an animal’s neural code over days with fixed behavior, is now documented across sensory, motor, and association cortex and in the hippocampus [14, 2, 16]. This has a stark impact on neural decoding: a decoder trained on an animal today degrades on that same animal weeks later. The dominant response is recalibration: periodically refitting the decoder on fresh, within-subject data [1, 4]. Recalibration requires ongoing labelled data from the individual, and it treats each subject in isolation.

We ask whether the code’s cross-individual structure offers an alternative route to temporal robustness. Cortex-wide behavioral representations are known to be considerably shared among individuals [9, 6, 7], which allows models to decode behavior from unseen individuals [17]. The question we pose is temporal: does a decoder built from a population of other individuals resist drift better than a decoder built from the subject itself? If individual brains drift along idiosyncratic, partly independent trajectories, then averaging over many individuals should cancel the individual-specific component of drift. This prediction, to our knowledge, has not been examined.

We test it on BraiDyn-BC [5], a widefield calcium-imaging resource whose ~ 15 -day operant protocol and shared 44-region Allen parcellation make within-subject and cross-subject decoders directly comparable across time. Our contributions:

- **Pooling resists drift, on two of four events.** Across two weeks a population-pooled decoder loses less accuracy than a mouse’s own (paired asymmetry $+0.017$ AUC, $p = 5 \times 10^{-5}$; Section 4.1). Per event the effect is significant for lick and reward, marginal for tone, and absent for lever pull, and a day-by-day regression over the whole protocol independently selects the same two events (Section 4.2).
- **It is not a data-volume artifact.** A count-matched control isolates the cause. At a fixed number of training events, a decoder trained on one other mouse ages less than one trained on the subject’s own, by $+0.007$ (lick), $+0.017$ (reward) and $+0.006$ AUC (tone). The gain therefore comes from training on individuals other than the subject, not from the larger training set pooling usually brings. Pooling over many mice rather than one lowers aging further on all four events, a smaller additional gain from a more diverse set of individuals (Sections 4.3–4.4).
- **Session-level validation is load-bearing.** Because trials within a recording session are not independent, a trial-level split inflates the personal decoder but not the pooled one, and so inflates the asymmetry from one side. We report the size of that inflation per event (Section 4.5); on lever pull it accounts for the entire apparent effect. We recommend session-held-out validation as the default for longitudinal decoding comparisons.
- **A drift-mitigation framing.** The early-week accuracy loss of cross-subject decoding is repaid as late-week stability, suggesting population priors as an alternative to per-subject recalibration. We release code and a no-raw-download streaming pipeline.

2 Related work

Representational drift and its mitigation. Drift is characterized within individual subjects [14, 2, 16, 13], and most approaches to mitigating it also operate within the subject. These include manifold stabilization [1], supervised and self-calibrating recalibration [4, 12], and adversarial or nonlinear alignment to the subject’s own reference day [3, 11]. Each method uses later data from the same subject to stabilize the decoder. We instead ask whether data from other subjects can provide that stability.

Cross-subject decoding. Shared structure across individuals already supports cross-subject readout through manifold alignment [8], contrastive embeddings [15], and, on this cohort, atlas-aligned foundation models [17]. Multi-participant pooling can also reduce overfitting: HTNet, for example, pools participant data to avoid fitting a decoder to a small single-subject sample [10]. Our question differs in two ways. First, our count-matched control keeps the training-set size fixed; the relevant overfitting is therefore not a consequence of limited data, but of relying on subject-specific features that later drift. Second, prior work measures accuracy or transfer at a fixed time. We measure how that accuracy changes over days, asking whether a pooled decoder ages more slowly.

Conservation of cortex-wide dynamics. Cortex-wide activity contains shared, low-dimensional, movement-related components [9, 6, 7]. Their presence makes cross-subject pooling possible, although it does not alone imply that pooling will resist drift.

3 Data and methods

Dataset. BraiDyn-BC [5] (DANDI:001425, CC-BY) provides, for each mouse and session, hemodynamically corrected dorsal-cortex $\Delta F/F$ traces parcellated into 44 Allen Common Coordinate

87 Framework regions at ~ 30 Hz, together with time-aligned behavioral channels. The dataset includes
 88 25 mice, each of which completed ~ 15 daily operant sessions over ~ 3 weeks. We stream the 44-
 89 region traces and behavioral channels without downloading the raw movies.

90 **Features and events.** We detect the onset of four events (lever pull, lick, reward, and tone) as rising
 91 edges, with a 1 s refractory period. For each onset we construct a 44-dimensional evoked feature
 92 by subtracting the mean pre-onset baseline (-0.5 – 0 s) from the mean post-onset $\Delta F/F$ response
 93 (0 – 1 s) in each parcel. Negative samples are drawn from matched-count windows more than 2 s
 94 from any event. Every classification is a binary decode of one event against these matched negatives,
 95 scored by area under the ROC curve (AUC).

96 **Decoders.** The primary decoder is a standardized ℓ_2 -regularized logistic regression on the 44-
 97 dimensional evoked feature. This is a linear readout of the parcel-space code: it recovers the compo-
 98 nent of behavior that is linearly present in the population average, and, because it has no capacity to
 99 memorize spatiotemporal detail, it cannot inflate an apparent generalization difference through over-
 100 fitting. It is also the natural comparison to prior cross-subject decoders, which are overwhelmingly
 101 linear [8, 10]. All conditions below use this same readout, so the only variable across conditions is
 102 which data the decoder was trained on.

103 **Temporal blocks and aging.** We group each mouse’s sessions into an early block (days 1–5) and a
 104 late block (days 11–15) and report AUC under four conditions. The personal decoder is estimated
 105 by leaving out one whole early *session* at a time: for each early session s we train on the mouse’s
 106 remaining early sessions and score the resulting decoder both on s (within-mouse same-block, WS)
 107 and on the late block (within-mouse across-block, WX), then average over s . Because WS and WX
 108 come from the same fitted decoders, they differ only in which data they are tested on. The pooled
 109 decoder is trained once on the other mice’s early blocks and scored on the same held-out sessions s
 110 (LS) and on the late block (LX).

111 Holding out sessions rather than trials is essential and not a detail. Trials within one recording share
 112 illumination and hemodynamic state, slow arousal drift, and temporal autocorrelation, so a trial-level
 113 split lets the personal decoder train and test inside the same session. The pooled decoder is trained
 114 on other animals entirely and cannot benefit this way, so the inflation is one-sided and lands directly
 115 on the quantity of interest. Section 4.5 reports what that costs per event.

116 Aging is the accuracy lost when testing moves from the held-out mouse’s early data to its late block
 117 while training stays fixed: personal aging is $WS - WX$ and pooled aging is $LS - LX$. The per-
 118 mouse aging asymmetry is $(WS - WX) - (LS - LX)$; a positive value means the personal decoder
 119 ages more. We assess significance with a paired sign-flip permutation test across mice (20,000
 120 permutations) and compute 95% confidence intervals with a cluster bootstrap over mice.

121 4 Results

122 4.1 A population-pooled decoder ages more slowly than a personal one

123 For each mouse, we keep the test data fixed (first the mouse’s held-out early sessions, then its late
 124 block) and change only the source of the training data: either the mouse itself or the other 24 mice.
 125 The pooled decoder loses less accuracy across the two weeks (Table 1). Across all 100 mouse–target
 126 pairs the asymmetry is $+0.017$ AUC ($p = 5 \times 10^{-5}$), with the personal decoder aging more in 73%
 127 of pairs.

128 The effect is not uniform across events, and we do not claim that it is. It is clearly present for reward
 129 ($+0.029$, $p = 0.0005$) and lick ($+0.020$, $p = 0.002$), marginal for tone ($+0.015$, $p = 0.08$), and
 130 absent for lever pull ($+0.003$, $p = 0.37$). Lever pull is the one event where both decoders *improve*
 131 across the protocol as the animal becomes more practiced, and once sessions are held out properly
 132 the pooled decoder’s improvement no longer exceeds the personal decoder’s by a detectable margin.
 133 Section 4.2 shows that a completely different estimator – a day-by-day regression rather than a two-
 134 block contrast – selects the same two events, which is the main reason we regard the lick and reward
 135 results as real rather than as a surviving tail of the same multiple-comparisons pool.

Table 1: Personal vs. pooled aging across days (AUC lost from early to late test, holding training fixed), per event, $n = 25$ mice, with whole early sessions held out. Asymmetry = personal – pooled aging; a positive value means the mouse’s own decoder ages more. p : paired sign-flip permutation. The rightmost column gives the same asymmetry under a trial-level split, the protocol this analysis replaces; the difference is the inflation quantified in Section 4.5.

Event	Personal	Pooled	Asymmetry (95% CI)	p	own ages more	trial-split
Lever-pull	−0.022	−0.025	+0.003 [−0.004, 0.011]	0.37	64%	+0.014
Lick	+0.012	−0.009	+0.020 [0.008, 0.032]	0.0023	80%	+0.028
Reward	+0.017	−0.012	+0.029 [0.014, 0.046]	0.0005	80%	+0.032
Tone	+0.016	+0.001	+0.015 [0.001, 0.032]	0.077	68%	+0.022
Pooled			+0.017 [0.010, 0.024]	5×10^{-5}	73%	+0.024

Decoding across the 15-day protocol: the personal decoder starts ahead, the pooled decoder gains faster, so the gap narrows

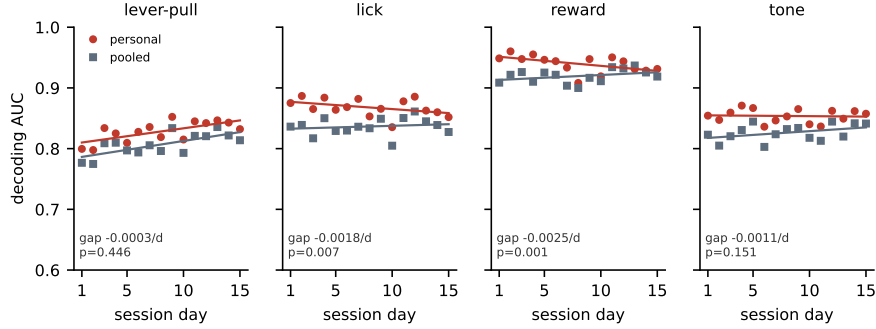


Figure 1: Decoding accuracy across the 15-day protocol for the personal decoder (own early data) and the pooled decoder (other mice’s early data), with per-day points and linear fits. The personal decoder starts ahead, and the pooled decoder gains more over days, so the gap between them narrows. The narrowing is significant for lick and reward, the two events where the personal decoder also loses accuracy over days, and is not significant for lever pull or tone. The per-event gap slope and its p (paired sign-flip over mice) are shown.

136 4.2 The personal decoder’s advantage erodes over the protocol

137 The block split compares two windows, but the effect should also appear as a continuous trend
138 across all fifteen days. For each session day we compute the accuracy of the personal decoder and
139 the pooled decoder and regress each on day; on early days the personal decoder is trained by leave-
140 one-session-out, so it is never tested on a session it trained on. The personal decoder starts ahead:
141 on day 1 it leads the pooled decoder by about 0.035 AUC, as expected for a decoder trained on
142 the same animal. That lead then erodes. The gap narrows by 0.0014 AUC per day, pooled over
143 all 100 mouse–target pairs ($p = 5 \times 10^{-5}$; the gap narrows in 65% of pairs), so by day 15 it has
144 fallen to about 0.014 AUC: roughly 60% of the early-week advantage is gone. Averaged over the
145 four events, the narrowing comes from the pooled decoder gaining accuracy over days (+0.0015
146 AUC/day) while the personal decoder stays flat (+0.0000 AUC/day); it is not that the personal
147 decoder collapses. Per event, the personal decoder does lose accuracy for lick (−0.0010 AUC/day)
148 and reward (−0.0016 AUC/day), and these are the two events where the narrowing is individually
149 significant ($p = 0.007$ and $p = 0.001$). For lever pull both decoders improve with practice, and for
150 tone the personal decoder is flat; in neither case is the gap slope significant (Fig. 1).

151 4.3 The robustness is not a data-volume artifact

152 A pooled decoder contains $\sim 24\times$ more training events than a personal decoder. Its stability could
153 therefore reflect data volume rather than pooling. We test this with a count-matched control (Fig. 2,
154 left). For each held-out early session s of each mouse we set the training-set size to n , the number of
155 events in that mouse’s remaining early sessions, and compare three decoders: *self*, trained on those

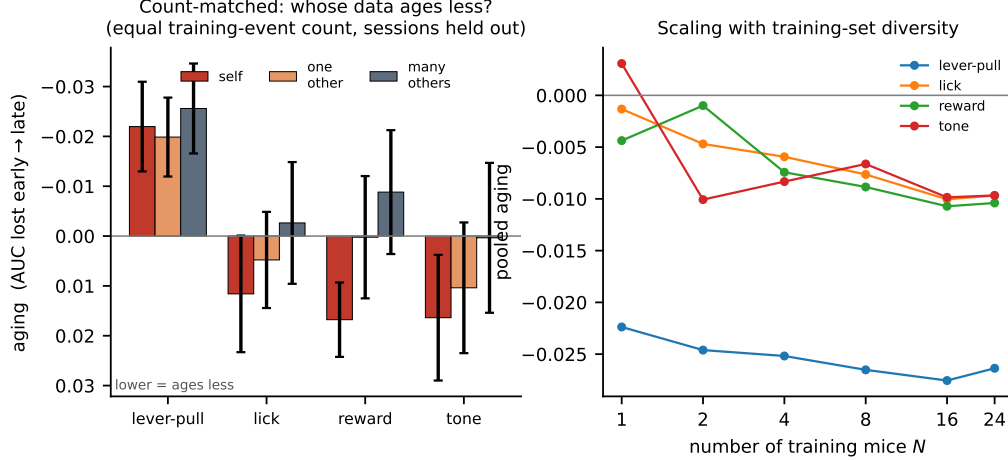


Figure 2: **Left:** count-matched control. With equal numbers of training events, decoder aging (AUC lost from early to late; lower values indicate greater robustness) is compared for the subject’s own data (self), one other mouse, and many other mice. The self decoder ages, whereas the other-individual decoders do not. **Right:** pooled aging as a function of the number of training mice N . Greater diversity provides a small additional gain, with immediate saturation for lever pull.

156 n events; *one*, trained on n events from a single other mouse; and *many*, trained on n events drawn
 157 across the other 24 mice. All three are scored on the same held-out session s and the same late block,
 158 so they differ only in whose data they were fit on.

159 For the events that showed an asymmetry in Section 4.1, the count-matched ordering reproduces
 160 it. The self decoder ages more than a decoder trained on one other mouse by +0.007 AUC for
 161 lick (+0.012 vs. +0.005), +0.017 for reward (+0.017 vs. +0.000), and +0.006 for tone (+0.016
 162 vs. +0.010). Because both decoders see the same number of training events, data volume cannot
 163 explain these gaps: what distinguishes them is only whether the training data came from the subject.
 164 The reading we favour is that the mouse’s own early decoder fits idiosyncratic features that later
 165 drift, features a decoder trained on another animal never had access to.

166 Lever pull again behaves differently: at matched count the self decoder ages *less* than the one-other
 167 decoder (−0.022 vs. −0.020), reversing the ordering. Consistently with Section 4.1, we do not
 168 claim the effect for this event.

169 The diversity ordering is the most robust result in this analysis: *many* ages less than *one* for all four
 170 events, including lever pull (−0.026 vs. −0.020; −0.003 vs. +0.005 for lick, −0.009 vs. +0.000
 171 for reward, +0.000 vs. +0.010 for tone). Training on a diverse set of individuals rather than a single
 172 other individual therefore provides a smaller, additional benefit that does not depend on which event
 173 is decoded.

174 4.4 Diversity: a secondary scaling with the number of training individuals

175 We next vary the number of training mice, $N \in \{1, 2, 4, 8, 16, 24\}$ (Fig. 2, right). As N increases,
 176 pooled aging becomes slightly more favorable for lick (−0.001 to −0.010), reward (−0.004 to
 177 −0.010), and tone (+0.003 to −0.010). Lever-pull performance saturates with a single training
 178 mouse and remains near −0.025. The scaling effect is around 0.01 AUC, comparable to the 0.006–
 179 0.017 self-versus-one-other gap of Section 4.3 rather than small beside it. Most of the resistance to
 180 drift is already present when training on a single other individual, and additional individuals refine it
 181 further; this analysis never lets the decoder see the held-out mouse, so it is unaffected by the protocol
 182 issue of Section 4.5.

183 4.5 How much the validation protocol matters

184 Every number above holds out whole sessions. Repeating the identical analysis on the identical fea-
185 tures with a trial-level split – shuffling trials pooled across the early sessions, so that a decoder may
186 train on one part of a session and be tested on another – inflates the personal decoder’s within-block
187 accuracy by +0.013 (lever pull), +0.010 (lick), +0.009 (reward) and +0.000 AUC (tone). The
188 pooled decoder is unaffected, because it is trained on other animals and has no session in common
189 with the test data.

190 The consequence is that the inflation lands on one side of a difference between two arms. Asymme-
191 tries measured this way are +0.014, +0.028, +0.032 and +0.022 for lever pull, lick, reward and
192 tone (rightmost column of Table 1), against +0.003, +0.020, +0.029 and +0.015 with sessions
193 held out. For lever pull that difference is the entire effect: what looks like a significant +0.014
194 ($p = 0.001$) becomes +0.003 ($p = 0.37$). For reward, which has the largest true effect, it changes
195 little.

196 The same reasoning applies to the count-matched control, where the natural implementation is to
197 score the self decoder on the block it was fit on. That is not a cross-validated quantity at all, and it
198 moves the self arm by -0.024 AUC on lever pull – enough to reverse the ordering the control exists
199 to establish.

200 We report this because the pattern generalizes beyond our dataset. Any comparison of a within-
201 subject decoder against a cross-subject one is exposed to it, since only the within-subject arm can
202 share a session between train and test. Where trials are not independent – which for physiolog-
203 ical recordings is the normal case – the split must respect the recording session, or the resulting
204 asymmetry will be biased in a predictable direction.

205 5 Discussion

206 A subject’s own decoder overfits idiosyncratic features of its early sessions, some of which later
207 drift. A decoder that has never seen the subject must instead rely on components of the code that
208 are shared across individuals and more stable over time. The count-matched control supports this
209 reading directly: at equal data, other-individual data ages less than the subject’s own for the events
210 where we detect the effect. Population structure is therefore not only a resource for cross-subject
211 transfer, but also for temporal robustness. The early-week accuracy cost of cross-subject decoding is
212 recovered over time as the pooled decoder remains stable. Unlike within-subject recalibration, this
213 approach does not require newly labelled data from the individual at test time.

214 The two events that carry the effect, lick and reward, are also the two whose personal decoders lose
215 accuracy over the protocol; the two that do not, lever pull and tone, are the two where the animal’s
216 performance is still improving and both decoders gain. This is consistent with the account above –
217 there is more subject-specific drift to avoid overfitting where drift is actually occurring – but we did
218 not predict the split in advance and treat it as an observation rather than a tested hypothesis.

219 **Limitations.** The effect is present for two of the four events we decode, marginal for a third, and
220 absent for the fourth; the pooled estimate over all 100 mouse–target pairs should be read with that
221 in mind rather than as a uniform property of cortex. We measure at the mesoscale, using parcel-
222 averaged activity, so whether the same result holds for single-neuron codes – where representational
223 drift is best characterized – remains unknown. The analysis covers a two-week interval, discrete
224 task events, and one species; longer horizons, graded behaviors, other species, and other recording
225 modalities should be tested. Our conclusions depend on within-versus-pooled and early-versus-late
226 comparisons rather than on absolute AUC. Although the count-matched control rules out training-
227 set size and is consistent with self-overfitting, it does not identify which components of drift are
228 individual-specific. Finally, we report only linear readouts here; whether a higher-capacity decoder
229 changes the picture is a question our protocol makes harder rather than easier to answer, since such
230 a model has more capacity to exploit exactly the within-session structure Section 4.5 shows must be
231 excluded.

Reproducibility

BraiDyn-BC is publicly available through DANDI:001425. We release the analysis code and a streaming feature pipeline that reproduces all reported results from the parcellated traces without downloading the raw imaging data. Section 3 and the released code specify all data splits, permutation seeds, and hyperparameters.

References

- [1] A. D. Degenhart et al. Stabilization of a brain–computer interface via the alignment of low-dimensional spaces of neural activity. *Nature Biomedical Engineering*, 4:672–685, 2020.
- [2] L. N. Driscoll et al. Dynamic reorganization of neuronal activity patterns in parietal cortex. *Cell*, 170(5):986–999, 2017.
- [3] A. Farshchian et al. Adversarial domain adaptation for stable brain–machine interfaces. *ICLR*, 2019.
- [4] B. Jarosiewicz et al. Virtual typing by people with tetraplegia using a self-calibrating intracortical brain–computer interface. *Science Translational Medicine*, 7(313):313ra179, 2015.
- [5] A. Kondo et al. BraiDyn-BC: a cortex-wide widefield calcium imaging dataset during a cued lever-pull operant task. *Scientific Data*, 12, 2025. DANDI:001425.
- [6] C. J. MacDowell and T. J. Buschman. Low-dimensional spatiotemporal dynamics underlie cortex-wide neural activity. *Current Biology*, 30(14):2665–2680, 2020.
- [7] C. J. MacDowell et al. Shared cortex-wide spatiotemporal motifs govern activity across individuals and behavioral states. *Current Biology*, 34, 2024.
- [8] S. Melbaum et al. Conserved structures of neural activity in sensorimotor cortex of freely moving rats enable cross-subject decoding. *Nature Communications*, 13:7902, 2022.
- [9] S. Musall et al. Single-trial neural dynamics are dominated by richly varied movements. *Nature Neuroscience*, 22(10):1677–1686, 2019.
- [10] S. M. Peterson, Z. Steine-Hanson, N. Davis, R. P. N. Rao, and B. W. Brunton. Generalized neural decoders for transfer learning across participants and recording modalities. *Journal of Neural Engineering*, 18:026014, 2021.
- [11] B. M. Karpowicz, Y. H. Ali, L. N. Wimalasena, et al. Stabilizing brain–computer interfaces through alignment of latent dynamics (NoMAD). *Nature Communications*, 16:4662, 2025.
- [12] G. H. Wilson et al. Long-term unsupervised recalibration of cursor-based intracortical brain–computer interfaces using a hidden Markov model. *Nature Biomedical Engineering*, 2025.
- [13] M. E. Rule, T. O’Leary, and C. D. Harvey. Causes and consequences of representational drift. *Current Opinion in Neurobiology*, 58:141–147, 2019.
- [14] M. E. Rule et al. Stable task information from an unstable neural population. *eLife*, 9:e51121, 2020.
- [15] S. Schneider, J. H. Lee, and M. W. Mathis. Learnable latent embeddings for joint behavioural and neural analysis (CEBRA). *Nature*, 617:360–368, 2023.
- [16] C. E. Schoonover et al. Representational drift in primary olfactory cortex. *Nature*, 594:541–546, 2021.
- [17] M. Hosseini, E. Erturk, S. Hashemi, and M. M. Shanechi. Cross-subject modeling for widefield calcium imaging via atlas-aligned spatiotemporal tokenization. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. arXiv:2607.09754.